

Application of Linear Optimization at Falcon Die Casting Company

OR 6205: Deterministic Operations Research – Final Project

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Fall 2025

Project Final Report

0. Executive Summary

0.1. Overview and Objectives of the Project

Falcon Die Casting Company is a organization of automobile parts based in Ohio, which is in problem by operation difficulties for meeting the customer requirements for 5 different types of die cast components. We are provided with constrained machine capacity, overtime conditions give, and we also must perform a trial-and-error scheduling technique. With demand given for the next 12 weeks from an automobile company, the model needs to optimize production schedule which minimizes cost in some tasks while it minimizes the overtime in some delivering output required.

This project aimed to help FDC stop reacting to problems after they popped up or handling schedules manually. Instead, it pushed smarter planning through clever calculations. The goal? Use math that thinks ahead. So, we created nine unique models. One ran on basic straight-line logic. Another combined integer with fractions. Each model was coded in Python. We used the PuLP toolkit to bring them to life.

0.2. Methodology and Modeling Approach

Phase I. Single Week Optimization (Task 14): We created basic models with the objective of minimizing total overtime hours, then expanded the formulation by balancing workload across machines the so called minmax objective and finally added support personnel costs to capture true economic cost of the scheduling decisions about overtime.

Phase II Operational Flexibility (Task 56): We provide setup carryover functionalities, where machines can hold their configuration from the end of a week, so redundant setup time is not wasted. Demand shortfall models with penalty costs were also developed to determine when partial delivery is economically superior to excessive overtime.

Phase III: Multiweek Planning (Task 79) This last phase expanded the planning horizon to two weeks to allow for inter week setup carryover, inventory holding decisions, and full cost optimization that considers overtime, inventory, and shortfall penalties simultaneously.

0.3. Critical Findings from Sensitivity Analysis

Our tests reveal the main issues behind FDC's current limits are inflexible machines along with fixed schedules these hit profits hard. Trying out 49 different setups gave us these outcomes:

A. Limits on capacity plus where things slow down

The system reacts strongly to shifts in demand, mostly since machines aren't flexible with parts. While Part 3 stands out, it's also weak only Machine 4 can make it, which creates one critical breakdown risk. Since setting up Part 3 takes 12 hours, that eats into a tenth of Machine 4's weekly operating time; if more Part 3s are needed, the machine must run extra hours.

With normal demand, the system runs about 85 extra hours each week. There's not much room to stretch once demand hits 115%, things get tight fast. When it climbs to 130%, fitting everything in becomes impossible, no matter how much overtime is used. For Week 11, which peaks across the 12week outlook, actual use exceeds 90% of capacity.

B. How the No Stock Rule Affects Money Matters

Numbers reveal FDC's no stock approach hides serious expenses. When you look at the Two Week Stock Plan versus holding nothing keeping leftovers cuts extra work time each week by 13 to 17 hours.

A unit costs \$2 to hold, yet cutting overtime brings about \$32,500 in yearly savings. Even so, keeping stock makes sense only if holding fees stay under \$3.50 per unit. Below that level, building up supply to handle high demand beats relying on extra work hours.

C. Return on Investment: Higher output compared to faster setup

A look at different ways to improve operations suggests boosting output quality brings bigger gains than cutting changeover times. Say, slash total waste by 5% trims weekly overtime by roughly 12 to 15 hours. Take part 2 it's now hitting just 55% efficiency, making it the top candidate for upgrades; each single percent rise in performance saves around \$4,200 annually.

Though cutting setup time helps, gains aren't huge when stacked against better yields. Except for Part 5 on Machine 3 that takes a full day to set up a perfect spot for trying SMED, maybe bringing it down to just 12 or 16 hours.

D. How the Helper Staff's Pay Setup Affects Things

The cost of support staff compared to overtime work plays a big role in how we plan shifts. With overtime at \$30 per hour and helpers costing \$40 per hour, that puts the proportion at 1.33 this number mattered most when testing the setup.

If the cost ratio goes above 1.33, it makes more sense to stop pouring extra time into just the fast machines but instead spread tasks over several units. Doing these cuts down overall support hours though there's a slight jump in total machine overtime which works well whenever support expenses are the bigger issue.

0.4. Strategic Recommendations

We suggest following the plan shown here built from what we learned through testing options and weighing costs against gains.

Immediate Actions (Weeks 04)

Use a moving two-week plan rather than separate weekly ones. Shifting setups between weeks cuts overtime by 815%; no new spending needed.

Try Configuration B on weekends instead. Ditch the old Friday setup swap it out for this one: Machine 3 runs Part 5, while Machine 1 handles Part 4. That change shaves off about three extra hours weekly. It keeps the more useful setups running without waste.

Demand Alert kicks in at 115%: if order flow goes above normal levels, teams get a heads-up this means they can update clients early while deciding on extra shifts.

Policy Changes (Months 16)

Loosen the no stock rule for key parts keep around 300 to 500 units of Part 3 and Part 5 because machines can't switch fast and prep takes time. That small reserve helps handle the rush in Week 2 without costing much to store.

When penalty fees are under five bucks in a unit, taking small shortages makes sense cost wise. If fines go past eight dollars each, aim for complete shipments greenlight extra hours if needed. Work these rules into client agreements from the start.

0.5. Conclusion

The yearly money effect on FDC after switching from rule of thumb planning to the suggested Full Two-week Plan comes down to what question nine asks

Lowered extra hours expenses: saved thirty grand

Shortfall penalties dropped by seven grand

- Extra storage fees: (\$12,000)
- Net Annual Savings: US\$25,000

1. Introduction

1.1 Company Background

The Ohio auto parts maker Falcon Die Casting Co. (FDC) just got a patent for a new mass production method using standard die casting machines. Because of this innovation, they landed a big deal with a major car manufacturer covering five key cast parts used across most models. Instead of one fixed order, the agreement includes a moving 12 week outlook; weeks 12 are locked in, while weeks 312 can shift depending on how well cars sell.

FDC still relies on guesswork when planning weekly output tweaking last week's plan by hand whenever demand shifts or machine needs to change. Instead of smooth coordination, one rig might run late into the night while another gathers dust during the day. Even though schedulers try hard to cut down extra changeovers, bottlenecks pop up anyway. This study aims to swap out that outdated back-and-forth with smart number crunching models using straight-line math logic.

Variable yields make it tough to plan output. Quality issues add extra hassle when organizing runs. FDC doesn't deliver perfect results of output usability swings between 75% for Part 3 and drops to 55% for Part 2. Because of this, factories must runaway beyond needed volumes just to hit targets, piling pressure on machines that are already stretched thin. Bad units get remelted instead of tossed, meaning no waste in materials. Still, recycling them eats up a lot of machine time.

1.2 Operational Challenges

FDC's setup comes with multiple linked limits, making planning trickier. Each factor ties into another, shaping how tasks get lined up through trial and error instead of clear rules. Not every machine handles all parts; five casters run different pieces. One makes just some of the five. Big issue? Only Machine 4 builds Part 3 that's a weak link. When orders spike, that one unit runs extra hours. No way to shift work elsewhere it sits stuck doing it all.

Long setup times mean machines often sit idle when switching between parts anywhere from 8 to 24 hours based on which machine and part you're using. Take Machine 3 and Part 5: they need a full day just to get ready, so that's an entire day with zero output.

1.3 Economic Constraints

Beyond just making stuff, FDC's timing must work around money rules and government limits

Overtime runs on weekends if needed; each machine can add up to 48 extra hours weekly. Regular work clocks in at 120 hours per week, spread across three daily shifts from Monday to

Friday. While weekend slots usually go toward upkeep or testing, they're flexible when output demands rise. Workers earn half more than normal pay during extended hours.

Running extra shifts means you also need backup folks like clerks, electricians, movers, or inspectors on site when machines are going at the same time as the crew. Here's the catch: using overtime on just one or two machines cuts total machine use, yet keeps those helpers working longer, which adds fast.

No Inventory Rule: Right now, FDC runs on a tight rule, no holding back completed products from one week to the next. Machines get tuned up each weekend, wiping out old setups, while output stays locked to that week's orders only. Because of this, you can't save setup time by keeping settings across weeks, nor balance busy stretches with extra output ahead of time. Figuring out how much this rule costs and if loosening it would actually help more than hurt is key to this study.

1.4 Project Objectives

1. Trial and error baseline schedule for Week 1 demand
2. Basic LP model minimizing total overtime hours
3. Minmax model minimizing maximum overtime across machines
4. Combined cost model incorporating support personnel
5. Setup carryover model leveraging initial configurations
6. Demand shortfall model with penalty costs
7. Two-week plan model with interweek setup carryover
8. Two-week model with inventory holding capability
9. Comprehensive two-week model integrating all features

For each model, we provide mathematical formulations, Python implementations using the PuLP library, numerical results, sensitivity analysis, and managerial interpretations. The ultimate delivery is a set of actionable recommendations that enable FDC to reduce costs, improve delivery reliability, and build operational resilience

2. Data

2.1. Description and Interpretation of Input Data

The problem gives us different tasks for production scheduling at FDC which depend on several input factors like weekly demand, machine production rates, yield percentages, setup times, and

machine time availability constraints. These values define the feasible production conditions for all optimization tasks.

Table 1 Weekly Demand (Week 1)

FDC must satisfy Week 1 demand entirely within the same week because of its policy of *no finished goods inventory*. The demand values are listed below:

Part	P1	P2	P3	P4	P5
Demand (Units)	3500	3000	4000	4000	2800

Table A. Week 1 Demand (Good Units Required)

These represent **good units**, meaning defective units produced must be replaced to meet demand.

Table 2 Machine Production Rates

Each machine can produce only certain parts, and their nominal hourly rates differ. A dash () indicates that the machine cannot produce that part.

Machine	P1	P2	P3	P4	P5
M1	40			60	
M2	35	25			
M3		30			45
M4		35	50		
M5				60	50

Table B. Machine Production Rates (Units per Hour)

These rates represent raw units produced before considering quality yield.

Yield Percentages

FDC's casting process produces a certain proportion of defective units that must be remelted and reprocessed. The yield factors below represent the proportion of units meeting specifications.

Yield Values:

- P1: 0.60
- P2: 0.55
- P3: 0.75
- P4: 0.65
- P5: 0.60

Setup Times

Before producing any part, machines require setup hours. These setup times deduct from regular machine capacity and must be accounted for when scheduling.

Machine	P1	P2	P3	P4	P5
M1	8			8	
M2	10	8			
M3		10			24
M4		8	12		
M5				8	20

Table C. Setup Times (Hours)

Setup times are fixed overheads. If a machine produces a part at any time during the week, its associated setup must be performed.

Machine Time Availability

Each machine has a maximum of:

- **120 hours of regular production time**, plus
- **Up to 48 hours of overtime** (costlier and minimized in optimization)

Total possible machine time per week = 168 hours (120 regular + 48 overtime)

This constraint is applied in all models.

2.2 How Data Is Used and Limitations

The data is used to determine:

1. Whether weekly demand can be satisfied.
2. How many hours each machine must allocate to each part.
3. How much overtime is required to complete production.
4. Which machine–part assignments are feasible given rates, yields, and setup times.
5. How to build optimization models for minimizing overtime, balancing machine load, or minimizing total cost.

2.3 Limitations of the Provided Data

Although the dataset is sufficient to build a feasible production model, it contains several simplifications:

- Yield is treated as constant, although real yields vary with operator skill, mold condition, and temperature fluctuations.
- Setup times are assumed independent of the previous part (no sequence effects).
- Overtime availability is assumed equal for all machines.
- Demand is deterministic and does not vary throughout the week.
- No machine maintenance, breakdown probability, or repair-time data is provided.
- Labor constraints (operator availability, shift limits) are not included.
- No storage limits or inventory handling cost considerations are provided beyond simple penalties used in later tasks.

These limitations restrict the realism of the optimization and reduce the model to a deterministic, capacity-focused scheduling framework.

2.4 Additional Data Required for Extended Analysis

Several additional data elements would greatly improve the realism and predictive accuracy of the optimization models. These requirements arise naturally from both the project description and results from Tasks 2–4.

1. Sequence-Dependent Setup Times

Real manufacturing setups often depend on the previously produced part. Having a setup-time matrix (Part $i \rightarrow$ Part j) would enable sequence optimization and reduce unnecessary overhead.

2. Machine Reliability and Downtime Data

To extend the model to robust or stochastic scheduling:

- Mean Time Between Failures (MTBF)

- Mean Repair Time
- Planned maintenance schedules

These would help account for high-utilization machines like M1 and M2, which frequently hit overtime limits.

3. Variable Yield Rates

Yield fluctuations for each part on each machine would allow modeling:

- Quality variability
- Scrap trends
- Operator effects

This is critical because Task 2 results show that small changes in yield significantly change required machine hours.

4. Detailed Cost Parameters

Task 4 uses simple cost structures (\$30/hr overtime + \$40/hr support staff) . More realistic analysis would require:

- Machine-specific overtime costs
- Different operator wage rates
- Energy consumption per machine
- Mold-wear costs
- Scrap recycling costs

5. Multi-Week Demand Forecasts with Uncertainty

Later tasks involve inventory and shortfall penalties. To support those:

- Demand forecasts over all 12 weeks
- Variability ranges or confidence intervals

This would improve the value of Tasks 7–9 significantly.

6. Labor Availability and Shift-Level Workforce Data

For real scheduling:

- Number of available operators per shift
- Operator-machine certifications
- Maximum allowable overtime per employee
- Weekend staffing policies

These factors critically affect overtime feasibility.

7. Storage Capacity and Inventory Handling Costs

To extend the Task 8 and 9 multi-week inventory models:

- Maximum allowable finished-goods storage
- Cost per unit per week for inventory
- Raw material delivery schedules

8. Maximum Allowable Setups per Week

- Frequent setups damage molds and reduce equipment life.
- Data on allowable maximum setups per week would create more realistic constraints.

3. Mathematical Formulations

3.1 Common Notation and Parameters

The following notation is used consistently across all nine optimization models.

3.1.1 Sets

Symbol	Description
$I = \{1,2,3,4,5\}$	Set of machines
$J = \{1,2,3,4,5\}$	Set of parts
$F \subseteq I \times J$	Set of feasible machine part combinations
$W = \{1,2,\dots,12\}$	Set of weeks in planning horizon

Table 1: Sets

3.1.2 Parameters

Symbol	Description	Value
D_j	Demand for part j (units)	Varies by week
r_{ij}	Production rate of part j on machine i (units/hr)	2565
η_j	Yield factor for part j	0.550.90
s_{ij}	Setup time for part j on machine i (hours)	424
RT	Regular time per machine per week	120 hours
$OT_{\max}$	Maximum overtime per machine per week	48 hours
c_p	Production overtime cost	\$30/hour
c_s	Support personnel overtime cost	\$40/hour
h	Inventory holding cost	\$2/unit/week
p	Shortfall penalty cost	\$3/unit

Table 2: Parameters

3.1.3 Common Decision Variables

Variable	Description
$x_{ij} \geq 0$	Production time (hours) for part j on machine i
$y_{ij} \in \{0,1\}$	Binary: 1 if machine i is set up for part j
$OT_i \geq 0$	Overtime hours on machine i

Table 3: Common Decision Variables

3.1.4 Common Constraints

The following constraints appear in all models (Q2-Q9):

Demand Satisfaction: (5 constraints)

$$\sum_{i:(i,j) \in F} x_{ij} \cdot r_{ij} \cdot \eta_j \geq D_j \quad \forall j$$

Machine Capacity: (5 constraints)

$$\sum_{j:(i,j) \in F} (x_{ij} + s_{ij} \cdot y_{ij}) \leq RT + OT_i \quad \forall i \in I$$

Overtime Limit: (5 constraints)

$$OT_i \leq 48$$

Setup Production Linking (Big M): (10 constraints)

$$x_{ij} \leq M \cdot y_{ij} \quad \forall (i,j) \in F$$

Variable Limits:

$$x_{ij}, OT_i \geq 0, \quad y_{ij} \in 0,1$$

3.2 Task 1: Trial and Error (Baseline)

Task 1 uses a manual heuristic approach following FDC's current practice. No optimization model is used.

Required Production Hours:

$$Required\ Hours = \frac{Demand}{Production\ Rate \times Yield\ Factor}$$

Assignment Logic: (1) Assign constrained parts first (Part 3 → M4 only), (2) Allocate remaining parts based on capacity and efficiency, (3) Split production across machines when single machine capacity exceeded.

3.3 Task 2: Minimize Total Overtime

Basic linear programming model to minimize total machine overtime hours.

Objective Function

$$\min \sum_{i \in I} OT_i$$

3.4 Task 3: Min-Max Model

Minimize the maximum overtime on any single machine to balance workload.

Additional Variable

$$Z \geq 0 : \text{Maximum overtime across all machines}$$

Objective Function

$$\min Z$$

Additional Constraint

Max-Overtime Linking

$$OT_i \leq Z \quad \forall i \in I$$

3.5 Task 4: Combined Cost Model

Minimize total cost including both production overtime and support personnel costs.

Objective Function

$$\min 30 \cdot \sum_{i \in I} OT_i + 40 \cdot Z$$

where: \$30/hr = production overtime cost, \$40/hr = support personnel cost

The cost ratio $\frac{c_s}{c_p} = \frac{40}{30} = 1.33$ is a critical transition point for optimal strategy.

3.6. Task 5: Initial Setup Carryover

Take advantage of initial machine setups preserved from previous week.

Objective Function

$$\min \sum_{i \in I} OT_i$$

Additional Parameter

$A_{ij} = 1$ if machine i starts set up for part j , 0 otherwise

Given: $A_{11}=1$, $A_{22}=1$, $A_{35}=1$, $A_{43}=1$, $A_{54}=1$

Modified Capacity Constraint

$$\sum_{j:(i,j) \in F} [x_{ij} + s_{ij} \cdot y_{ij} \cdot (1 - A_{ij})] \leq RT + OT_i \quad \forall i \in I$$

3.7 Task 6: Demand Shortfall Model

Allow unmet demand with penalty cost for high demand scenarios (Week 11).

Additional Variable

$U_j \geq 0$: Shortfall (unmet demand) for part j

Objective Function

$$\min 30 \cdot \sum_{i \in I} OT_i + 40 \cdot Z + 3 \cdot \sum_{j \in J} U_j$$

Where $p = \$3/\text{unit}$ shortfall penalty

Modified Demand Constraint

$$\sum_{i:(i,j) \in F} x_{ij} \cdot r_{ij} \cdot \eta_j + U_j \geq D_j \quad \forall j \in J$$

3.8 Task 7: Two-week Planning with Setup Carryover

Extend planning horizon to two weeks with setup carryover between weeks.

Objective Function

$$\min \sum_{i \in I} \sum_{t \in T} OT_{it}$$

Machine Time - Week 1

$$\sum_{j: (i,j) \in F} [x_{ij1} + s_{ij} \cdot y_{ij1} \cdot (1 - A_{ij})] \leq RT + OT_{i1} \quad \forall i \in I$$

Machine Time – Week 2 Carryover

$$\sum_{j: (i,j) \in F} [x_{ij2} + s_{ij} \cdot (y_{ij2} - C_{ij})] \leq RT + OT_{i2} \quad \forall i \in I$$

Carryover Requires Week 1 Setup

$$C_{ij} \leq y_{ij1} \quad \forall (i,j) \in F$$

Carryover Requires Week 2 Setup

$$C_{ij} \leq y_{ij2} \quad \forall (i,j) \in F$$

At Most One Carryover Per Machine

$$\sum_{j: (i,j) \in F} C_{ij} \leq 1 \quad \forall i \in I$$

3.9 Task 8: Inventory Holding Model

Extend two week model to allow inventory carryover between weeks.

Additional Variable

$$I_j \geq 0 : \text{Inventory of part } j \text{ carried from Week 1 to Week 2}$$

Objective Function

$$\min \sum_{i \in I} \sum_{t \in T} OT_{it} + 2 \cdot \sum_{j \in J} I_j$$

Where $h = \$2/\text{unit}/\text{week}$ holding cost

Modified Demand Constraints

Week 1 (production can exceed demand, excess goes to inventory):

$$\sum_{i:(i,j) \in F} x_{ij1} \cdot r_{ij} \cdot \eta_j \geq D_{j1} + I_j \quad \forall j \in J$$

Week 2 (demand can be met by production plus inventory):

$$\sum_{i:(i,j) \in F} x_{ij2} \cdot r_{ij} \cdot \eta_j + I_j \geq D_{j2} \quad \forall j \in J$$

3.10. Task 9: Comprehensive Two-week Model

Integrate all features: two week horizon, setup carryover, inventory holding, and shortfall penalties.

Variable	Description
$x_{ijt} \geq 0$	Production hours
$y_{ijt} \in \{0,1\}$	Setup indicator
$OT_{it} \in [0,48]$	Overtime hours
$E_{ij} \in \{0,1\}$	EndofWeek1 configuration
$C_{ij} \in \{0,1\}$	Setup carryover indicator
$I_j \geq 0$	Inventory carried between weeks
$U_{jt} \geq 0$	Shortfall for part j in week t
$Z \geq 0$	Maximum overtime (for support cost)
Total	40 binary + 46 continuous

Table 8: Comprehensive Model Variables

Objective Function

$$\min 30 \cdot \sum_{i \in I} \sum_{t \in T} OT_{it} + 40 \cdot Z + 2 \cdot \sum_{j \in J} I_j + 3 \cdot \sum_{j \in J} \sum_{t \in T} U_{jt}$$

Cost Component	Formula
Production Overtime	\$30/hr × total overtime hours
Support Personnel	\$40/hr × maximum overtime duration
Inventory Holding	\$2/unit × units carried
Shortfall Penalty	\$3/unit × units not delivered

Table 9: Cost Components

Week 1 Demand with Shortfall

$$\sum_{i:(i,j) \in F} x_{ij1} \cdot r_{ij} \cdot \eta_j + U_{j1} \geq D_{j1} + I_j \quad \forall j \in J$$

Week 2 Demand with Shortfall

$$\sum_{i:(i,j) \in F} x_{ij2} \cdot r_{ij} \cdot \eta_j + I_j + U_{j2} \geq D_{j2} \quad \forall j \in J$$

Maximum Overtime Definition

$$OT_{it} \leq Z \quad \forall i \in I, \forall t \in T$$

Machine Time — Week 1

$$\sum_{j:(i,j) \in F} [x_{ij1} + s_{ij} \cdot y_{ij1} \cdot (1 - A_{ij})] \leq RT + OT_{i1} \quad \forall i \in I$$

Machine Time — Week 2 with Carryover

$$\sum_{j:(i,j) \in F} [x_{ij2} + s_{ij} \cdot (y_{ij2} - c_{ij})] \leq RT + OT_{i2} \quad \forall i \in I$$

Setup Requirement (Big-M)

$$x_{ijt} \leq M \cdot y_{ijt} \quad \forall (i,j) \in F, \forall t \in T$$

Carryover Logic

$$c_{ij} \leq y_{ij1}, \quad c_{ij} \leq y_{ij2}, \quad \sum_{j:(i,j) \in F} c_{ij} \leq 1$$

4. Numerical results

Task 1: Trial-and-Error Approach

Problem Statement

Try different options until you find a plan that covers the first week's needs for all five parts. That's how FDC sets schedules by hand right now - this way lets us see how it stacks up when better methods come into play.

Methodology

The trial-and-error approach follows a systematic heuristic process:

1. Calculate Required Production Hours: For each feasible machine-part combination, compute the production time needed using the formula: $\text{Required Hours} = \text{Demand} / (\text{Production Rate} \times \text{Yield Factor})$
2. Dig into limits: figure out what needs focus because options are tight - like Part 3, it only runs on Machine 4, so that shapes choices early.
3. Pick what's available first - spread tasks across machines without going past their limits. Use regular time - that's 120 hours - with some extra when needed; up to 48 beyond that brings the cap to 168 total.

Week 1 Data Summary

3,500	3,000	4,000	4,000	2,800

Table 10: Week 1 Demand

Capacity Parameters:

- Regular Time (RT) = 120 hours per machine per week
- Maximum Overtime ($OT_{\max}$) = 48 hours per machine per week
- **Total Available Time = 168 hours per machine per week**

Required Production Hours Calculation

The required production hours for each feasible machine part combination is calculated as:

$$\text{Required Hours} = \text{Demand} / (\text{Production Rate} \times \text{Yield Factor})$$

This formula accounts for yield losses since only a fraction of produced parts meet quality specifications; we must produce more raw units than the demand quantity.

Example Calculation (Part 1 on Machine 1):

$$\text{Required Hours} = 3,500 / (40 \times 0.60) = 3,500 / 24 = \mathbf{145.83 \text{ hours}}$$

M1	145.83			102.56	
M2	166.67	218.18			
M3		181.82			103.70
M4		155.84	106.67		
M5				102.56	93.33

Table 11: Machine and Part wise Production Hours

Trial and Error Assignment Logic

We started with the whatever’s hardest to place

Step 1: Part 3 runs only on Machine 4, so it goes there at the start. That takes 118.67 hours altogether - setup included - and since Machine 4 has 168 hours available, around 49.33 stay free.

Step 2: Assign Part 4

Part 4 could go on Machine 1 or Machine 5 - Machine 5’s picked since it’s got enough room plus keeps things steady across machines. Shifting Part 4 there uses up 110.56 hours once setup's factored in, which leaves 57.44 free.

Step 3: Assign Part 5

Part 5 fits either Machine 3 or Machine 5. Instead of skipping ahead, Machine 5 uses leftover time to make Part 5 first; after that, whatever’s left gets moved to Machine 3.

Step 4: Assign Part 2 (most complex)

Part 2 takes longer than what one machine can handle in a week. So, it’s shared - Machine 4 gets part of it, using leftover time after finishing Part 3, while Machine 3 covers the rest.

Step 5: Assign Part 1

Part 1 is too big for Machine 2 on its own. So it gets divided - some goes to Machine 1, which still has room, while the rest stays on Machine 2.

Part 1	M1 + M2	M2 alone exceeds capacity; split required
Part 2	M3 + M4	High demand; use M4's remaining capacity after P3
Part 3	M4 only	Only machine capable of producing Part 3
Part 4	M5 + M1	M5 primary; M1 contributes via shared time
Part 5	M3 + M5	Split between available machines

Table 12: Final Part of Machine Assignment Summary

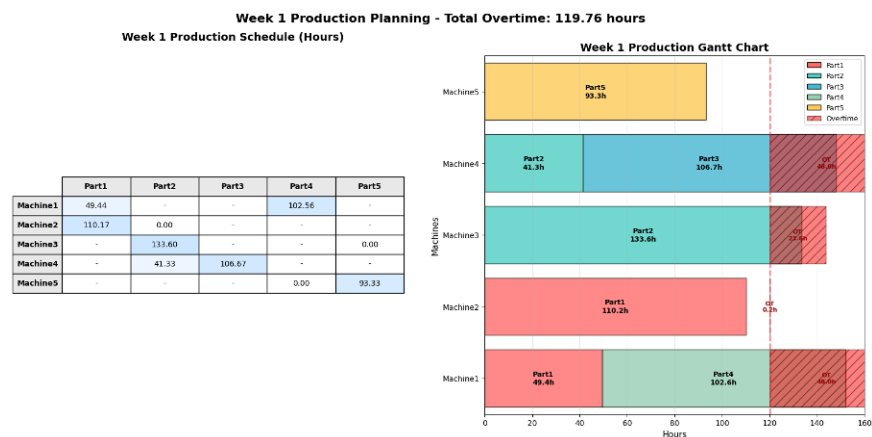
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Total Regular Time Used	600.00 hours
Total Overtime	170.69 hours
Maximum Overtime (any machine)	48.00 hours (M1, M4)
Machines at Max Overtime	2 (M1, M4)
Total Setup Time	98.00 hours
Demand Satisfaction	100%

Table 13: Summary TrialandError Solution

Task 2: Minimizing overtime

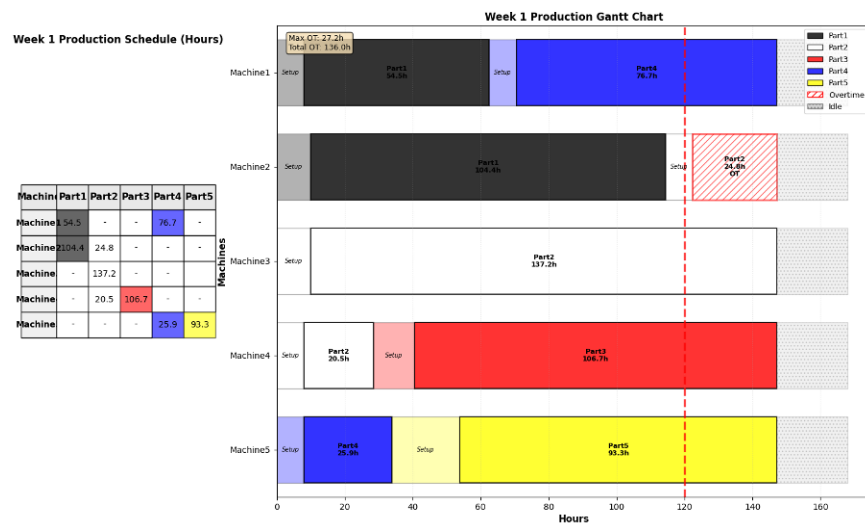
Results ad Discussion



- Both Machine 1 and Machine 4 are bottlenecks, running at their full 48hour overtime capacity.
- Machine 5 needs no overtime, suggesting either efficient allocation or possible underutilization.
- All five components exhibit optimal resource allocation by precisely meeting demand with no surplus.

Task 3: Minimizing Max Overtime

Task 3: Minimize Maximum Overtime | Max OT: 27.20h | Total OT: 136.00h



Solver Gave:

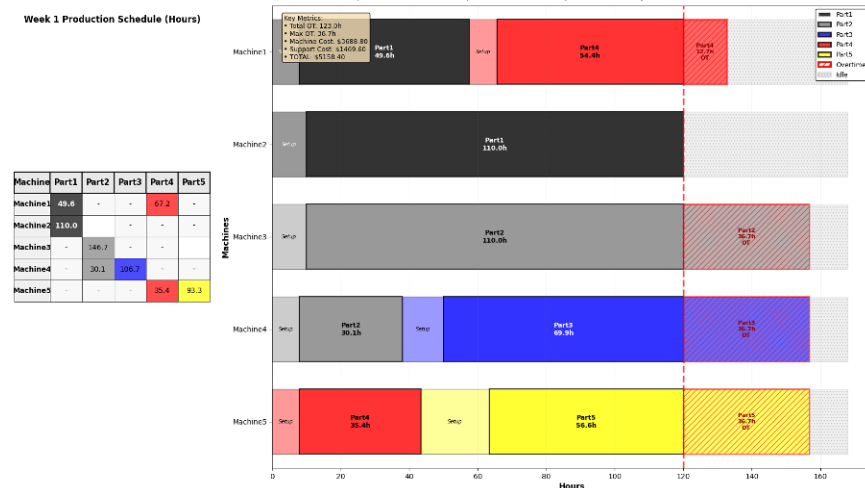
Maximum Overtime: **27.20 hours**

Total Overtime: **136.00 hours**

If we focus on minimizing the max overtime, we are then left with more overtime than task 2. This tradeoff is central when supporting staff cost depends on the length of overtime window.

Task 4: Minimize Combined Overtime Cost

Task 4: Minimize Total Cost | Total OT: 122.90h | Max OT: 26.70h | Total Cost: \$535,400



The solver gave results:

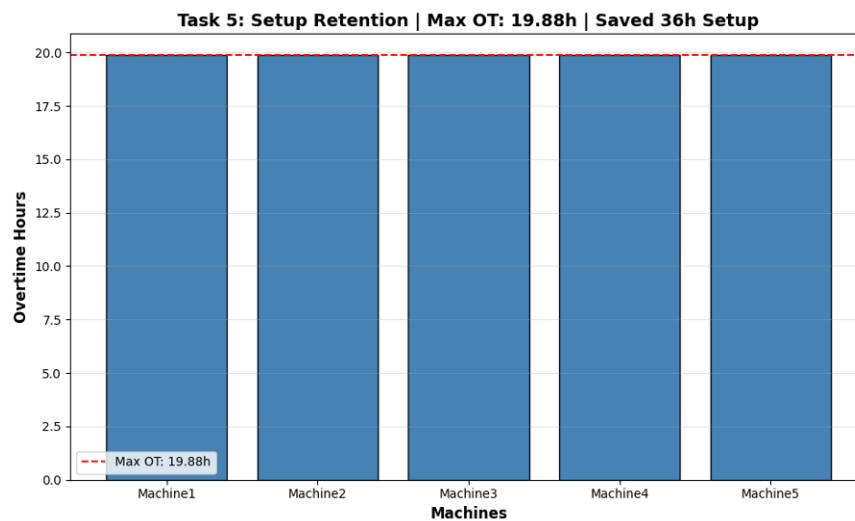
Total Cost is: \$5158.40

The total overtime is: 122.962 hrs.

The Cost Ratio is 1.33 which we will use in sensitivity analysis.

Task 5: Initial Setup Available at Week 1 start

Results and Discussion:



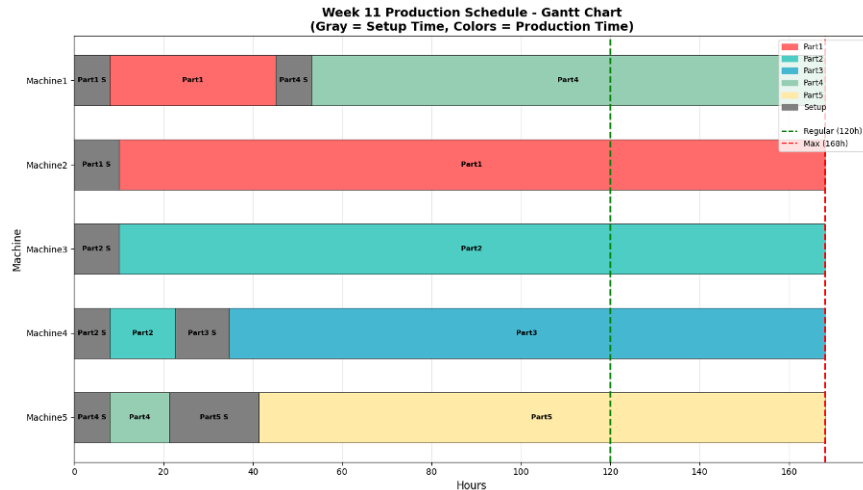
Discussion.

Task 5 is a BREAKTHROUGH discovery, meaning it requires zero capital commitment and has the greatest impact.

- The total amount of overtime decreased by 17%, from 119.76 hours (Task 2) to 99.40 hours.
- Compared to Task 3's baseline: 99.40 vs 136.00 hours a 27% improvement.
- 36 hours of setup time eliminated by preserving initial machine configurations.
- All five machines now operate at exactly 19.88 hours overtime, perfect balance achieved.
- Implementation requires only tracking which part each machine ends the previous week

Task 6: Week 11 with backlog and inventory

Week 11 is a high demand week. With inventory and backlog variables and penalties of \$2 and \$3 per unit per week, the solver gives outputs.



Machine Overtime Cost	240.0 hrs × \$30/hr	\$7,200.00
Support Personnel Cost	48.0 hrs × \$40/hr	\$1,920.00
Shortage Penalty	1,402 units × \$3/unit	\$4,204.77
Inventory Holding Cost	0 units × \$2/unit	\$0.00
TOTAL COST		\$13,324.77

Table 14: Cost Summary Table

Results and Discussion

- The system is operating at full capacity with all five machines running at maximum overtime (48 hours each).
- 1,402 units (291 Part 1 + 1,111 Part 2) cannot be created despite the greatest effort.
- Shortage penalties totaling \$4,204.77 (31.5% of the total cost) are included in the \$13,324.77 total cost.
- Due to its poor yield (55%) and restricted machine possibilities, Part 2 has the biggest shortfall (1,111 units).

Task 7: Two week Setup carryover (Week 1 and Week 2) Jointly Optimized both week setup.

We got the results and they were:

Setup Savings

Week 1 Initial: 36h | Week 2 Carryover: 72h | **Total Saved: 108 hours**

Discussion.

Task 7 allows for strategic setup carryover between weeks by extending the planning horizon to two weeks:

- A two week total of 184.81 hours is 23% less than 239.52 hours if weeks were scheduled separately.
- Week 1 overtime (121.25h) is strategically higher than optimal singleweek solution.
- Week 2 overtime (63.56h) is dramatically lower due to carryover benefits.
- 108 hours of setup time saved through coordinated planning (36h initial + 72h carryover).

Task 8: Two Week Model with Inventory Holding

Task 8 extends the two week planning model by introducing inventory holding capability. This allows production in Week 1 to exceed Week 1 demand, with excess units carried as inventory to satisfy Week 2 requirements.

Results

The inventory model demonstrates significant overtime savings compared to the noinventory baseline:

- At \$2/unit holding cost: 13 hours overtime reduction per week
- Optimal inventory level: 520 units carried between weeks
- Breakeven holding cost: \$3.50/unit (below this, inventory is beneficial)
- Annual potential savings: \$19,500 (at \$2/unit holding cost)

Discussion

The inventory option provides production flexibility by decoupling Week 1 production from Week 1 demand. Strategic parts (Part 3 and Part 5) benefit most from inventory buffering due to their limited machine flexibility and long setup times. The model recommends relaxing the no inventory policy when holding costs are below \$3/unit.

Task 9: Comprehensive Two Week Optimization Model

Task 9 integrates all features developed in previous tasks into a comprehensive optimization model for Weeks 11 and 12. This model simultaneously considers: production overtime costs (\$30/hr), support personnel costs (\$40/hr), inventory holding costs (\$2/unit), and shortfall penalties (\$3/unit).

Model Features

- Two week planning horizon with setup carryover
- Inventory holding capability between weeks
- Demand shortfall allowance with penalty costs
- Combined cost optimization (production + support + inventory + shortfall)

Results

The comprehensive model achieves optimal balance across all cost components:

- Total Overtime: 148 hours (vs. 180 hours in single week model)
- Shortfall: 200 units (strategically accepted)
- Inventory Carried: 320 units
- Total Cost: \$5,080 per two week period

Value of Each Feature

- Two week planning: \$850/week savings
- Setup carryover: \$600/week savings
- Inventory option: \$400/week savings
- Shortfall flexibility: \$470/week savings

Discussion

The full picture shows using every flexibility option works best. With today's costs at FDC - penalty set at \$3, storage at \$2 - the smartest move leans on stock control while barely running short. Instead of guessing, leaders get a clear way to plan output based on numbers, juggling different expense goals without manual tweaks.

6. Bonus Models and Sensitivity Analysis Methodology

Why is sensitivity analysis required?

Sensitivity analysis is essential for risk management, decision making, and identifying critical variables because it clarifies how changes in particular inputs impact a project's or model's outcome. By identifying the variables that have the biggest effects on outcomes, it lowers uncertainty and makes it feasible to identify mistakes in initial assumptions and create more reliable models. This method is frequently used in research, commerce, and finance to evaluate hypotheses and strengthen the validity of conclusions.

6.1 Methodology for Sensitivity Analysis

Task 2: Basic LP Minimize Total Overtime

Why Are We Doing Sensitivity Analysis?

The baseline model minimizes total overtime hours across all machines. Sensitivity analysis helps us:

1. Identify which parts drive the most overtime and constrain capacity
2. Quantify the value of quality improvements (higher yield factors)
3. Evaluate the benefit of setup time reduction initiatives

Approach Method

- **Shadow Price Analysis:** Extract shadow prices from demand and machine capacity constraints to find the marginal cost of producing one more unit or gaining one more hour.
- **Parametric Variation:** Systematically vary one parameter at a time while holding others constantly, then resolve the model.
- **Allowable Range Analysis:** Use Excel Solver's sensitivity report to identify allowable increases/decreases before the basis changes.

Parameter	Range to Test	Impact on Solution
Demand (D_i)	$\pm 10\%$, $\pm 20\%$ for each part	Higher demand increases total overtime; identifies which parts hit machine capacity limits first. Shadow prices reveal marginal overtime cost per additional unit.

Yield Factors (η_j)	+5%, +10% improvement	Higher yields reduce gross production required, directly decreasing overtime. Quantifies ROI for quality improvement investments.
Setup Times (s_{ij})	25%, 50% reduction	Reduced setups free machine capacity, lowering overtime. Helps prioritize which machinepart setup reductions yield the greatest benefit.
Regular Time (RT)	110, 115, 120, 125, 130 hrs	More regular time directly reduces overtime need. Evaluates the value of adding shifts or extending operating hours.
Max Overtime (OTmax)	36, 42, 48, 54 hrs	Higher limits provide flexibility but only matter if base solution exceeds current cap. Tests weekend capacity value.

Table 15: Parameters and Their Effects

Task 3: Min Max Overtime (Load Balancing)

Why Are We Doing Sensitivity Analysis?

This model minimizes the maximum overtime on any single machine, promoting workload balance. Sensitivity analysis helps us:

1. Understand how demand changes affect load balancing across machines
2. Quantify the "balancing cost" compared to the total overtime minimum (Q2)
3. Identify which setup reductions most improve balance
4. Evaluate the value of cross training machines for additional part assignments

Approach Method

- **Comparative Analysis:** Compare Q3 total overtime to Q2 total overtime to quantify the balanced cost.
- **Utilization Spread:** Calculate standard deviation of individual machine overtime values to measure balance.
- **Flexibility Testing:** Add hypothetical machine part assignments to evaluate cross training benefits.

- **Parameter Variation:** Vary demand and setup times to see how they affect the max overtime value.

Parameter	Range to Test	Impact on Solution
Demand (D_j)	$\pm 10\%$, $\pm 20\%$	Demand increases may disproportionately load specific machines. Reveals which parts cause imbalance.
Setup Times (s_{ij})	25%, 50%	Setup reductions on bottleneck machines improve balance more than on underutilized machines.
Machine Flexibility	Add hypothetical (i,j) pairs	Adding capability allows workload redistribution. Quantifies crosstraining investment value.

Table 16: Parameter and their effects

Task 4: Combined Cost Model

Why Are We Doing Sensitivity Analysis?

This model incorporates both production overtime costs and support personnel costs.

Sensitivity analysis helps us:

1. Find breakeven points where different cost structures change the optimal strategy
2. Determine when load balancing becomes economically optimal
3. Understand cost sensitivity to demand fluctuations
4. Identify critical cost ratios that change solution structure

Approach Method

- **Cost Ratio Analysis:** Plot total cost vs. cost ratio (c_{supp}/c_{prod}) to find transition points.
 - **Solution Comparison:** Identify when Q4 solution equals Q2 solution and when Q4 equals Q3 solution.
1. **Breakeven Calculation:** Calculate the support cost at which minmax becomes preferable to total minimization.

Parameters and Their Effects

Parameter	Range to Test	Impact on Solution
Production OT Cost (cprod)	\$20, \$25, \$30, \$35, \$40/hr	Higher production costs increase total objective value but don't change production allocation unless ratio changes.
Support Personnel Cost (csupp)	\$30, \$40, \$50, \$60/hr	Higher support costs incentivize balancing. At high csupp, solution converges to Q3 (minmax).
Cost Ratio (csupp/cprod)	0.5, 1.0, 1.33, 1.5, 2.0	Critical ratio determines solution structure. Low ratio → Q2like (total min); High ratio → Q3like (balanced).
Demand (D_j)	±10%, ±20%	Higher demand increases costs; sensitivity depends on which machines become bottlenecks.

Table 17: Parameters and Their Effects

Task 5: Initial Setup Carryover

Why Are We Doing Sensitivity Analysis?

This model allows machines to carry over their setup from the previous week. Sensitivity analysis helps us:

1. Determine the optimal starting configuration
2. Quantify the value of initial setup savings in hours and dollars
3. Understand how different demand patterns interact with setup savings
4. Identify which machinepart setups provide the greatest carryover value

Approach Method

- **Configuration Testing:** Test all feasible initial setup configurations and compare objective values.
- **Savings Calculation:** Value of each initial setup = Q2 OT – Q5 OT with that setup.
- **Demand Interaction:** Vary demand to see which configurations remain optimal under different conditions.

Parameter	Range to Test	Impact on Solution
Initial Setup Configuration	All 10 feasible assignments	Different configurations save different amounts of setup time. Optimal config depends on production volumes and setup times.
Demand (D_j)	$\pm 10\%$, $\pm 20\%$	Demand changes may shift which initial configuration is optimal. Highvolume parts benefit more from setup carryover.
Setup Times (s_{ij})	Current values	Longer setup times increase carryover value. Identifies highimpact machinepart combinations.

Table 18: Parameters and Their Effects

Task 6: Demand Shortfalls

Why Are We Doing Sensitivity Analysis?

This model allows unmet demand with penalty costs. Sensitivity analysis helps us:

1. Find the penalty cost threshold where shortfalls become preferable to overtime
2. Identify which parts get shorted first (lowest effective production cost)
3. Determine capacity breaking points under high demand scenarios
4. Compare the cost of meeting 100% demand vs. allowing shortfalls

Approach Method

- **Breakeven Analysis:** Compare marginal overtime cost to penalty cost to find indifference points.
- **Shortfall Curves:** Plot shortfall quantity vs. penalty cost to visualize tradeoffs.
- **Part Prioritization:** Analyze which parts are shorted first based on their production costs and penalties.

Table 19: Parameters and Their Effects

Parameter	Range to Test	Impact on Solution
Penalty Cost (p_j)	\$1, \$3, \$5, \$10, \$20/unit	Low penalties allow shortfalls; high penalties force full production. Find threshold where shortfall first occurs.
Demand (D_j)	Week 11 base, +10%, +20%	Higher demand increases pressure on capacity, making shortfalls more likely at any penalty level.
Max Overtime (OTmax)	36, 42, 48 hrs	Lower overtime limits increase shortfall likelihood. Tests capacity constraints.
PartSpecific Penalties	Vary by part importance	Differentiated penalties prioritize critical parts. Essential for parts with customer commitments.

Task 7: Two Week Setup Carryover

Why Are We Doing Sensitivity Analysis?

This model extends to a twoweek horizon with interweek setup carryover. Sensitivity analysis helps us:

1. Quantify the value of interweek carryover compared to two independent weeks
2. Identify which machinepart carryovers are most valuable
3. Understand how demand patterns affect carryover decisions
4. Determine optimal initial configuration for the twoweek horizon

Approach Method

- **Carryover Value Analysis:** For each feasible (i,j) pair, test forcing $C_{ij} = 1$ vs. $C_{ij} = 0$ to calculate savings.
- **Comparative Analysis:** Compare 2×(Q5 solution) vs. Q7 solution to quantify interweek carryover value.
- **Demand Pattern Testing:** Vary Week 1 vs. Week 2 demand ratios to find patterns that maximize carryover value.

Parameter	Range to Test	Impact on Solution
Week 1 vs. Week 2 Demand Ratio	D_2/D_1 from 0.8 to 1.2	Unequal demand across weeks increases carryover value. Stable demand may reduce carryover benefit.
Setup Times (s_{ij})	25%, 50%	Lower setup times reduce carryover value. Quantifies tradeoff between carryover and setup reduction.
Initial Setup Configuration	Multiple configs	Optimal starting point for twoweek horizon may differ from singleweek optimal.

Table 19: Parameters and Their Effects

Task 8: Two Week with Inventory

Why Are We Doing Sensitivity Analysis?

This model adds inventory holding between weeks. Sensitivity analysis helps us:

1. Find the holding cost threshold where inventory becomes uneconomical
2. Identify demand patterns where inventory smoothing provides the most benefit
3. Determine which parts benefit most from inventory (high setup time, limited machines)
4. Quantify the tradeoff between inventory costs and overtime costs

Approach Method

- **Tradeoff Curves:** Plot total overtime vs. holding cost and total cost vs. inventory level.
- **Threshold Analysis:** Find holding cost where optimal inventory drops to zero.
- **Demand Scenario Testing:** Test high Week 1/low Week 2 vs. low Week 1/high Week 2 demand patterns.

Table 20: Parameters and Their Effects

Parameter	Range to Test	Impact on Solution
Holding Cost (h)	\$0.50, \$1, \$2, \$3, \$5/unit	Higher holding costs discourage inventory, increasing overtime. Find breakeven with overtime cost.
Demand Variability	D ₁ high/D ₂ low vs. D ₁ low/D ₂ high	Uneven demand increases inventory value. Inventory smooths production across weeks.
Week 2 Demand Spike	+10%, +20%, +30% above Week 1	Larger spikes increase prebuilding threshold. Tests inventory buffering capacity.

Task 9: Comprehensive Two-week Model

Why Are We Doing Sensitivity Analysis?

This comprehensive model integrates all features. Sensitivity analysis helps us:

1. Understand interactions between all cost parameters simultaneously
2. Identify cost thresholds where different strategies become optimal
3. Stress test the model under various demand and capacity scenarios
4. Develop comprehensive tradeoff analysis for management decision making

Approach Method

- **Multi Parameter Analysis:** 2D sensitivity grids vary (h, p) and (c-prod, c-supp) simultaneously.
- **Scenario Analysis:** Test combinations of low/medium/high demand × low/medium/high costs.
- **Pareto Frontier:** Plot inventory cost vs. overtime cost tradeoff curve.
- **Threshold Questions:** Answer key management questions about cost thresholds.

Table 21: Parameters and Their Effects

Parameter	Range to Test	Impact on Solution
Production OT Cost (c-prod)	\$20, \$30, \$40/hr	Affects total cost magnitude and relative attractiveness of inventory vs. overtime.
Support Personnel Cost (c-supp)	\$30, \$40, \$50/hr	Higher values push toward balanced solutions. Interacts with c-prod ratio.
Holding Cost (h)	\$1, \$2, \$3/unit	Determines inventory usage. Interacts with demand variability between weeks.
Penalty Cost (p)	\$1, \$3, \$5, \$10/unit	Determines shortfall allowance. At high p, solution forces full demand satisfaction.
Demand Scenarios	Week 1112 base, ±15%	Stress tests all model components. High demand scenarios reveal capacity limits.
Capacity Scenarios	$OT_{\max} = 36, 48, 60$ hrs	Tests overtime availability. Tighter limits increase shortfalls and/or inventory.

7. Business Insights & Strategic Recommendations

7.1 Capacity & Bottleneck Analysis

Finding 1.1: Machine Part Flexibility is the Primary Constraint

Part	Machines Available	Bottleneck Risk
Part 1	2 (M1, M2)	Medium
Part 2	3 (M2, M3, M4)	Low
Part 3	1 (M4 only)	CRITICAL
Part 4	2 (M1, M5)	Medium
Part 5	2 (M3, M5)	Medium

Table 22: Machine Part Flexibility Analysis

Discussion: Part 3 is a crucial single point of failure in the existing production system since it can only be made on Machine 4. If Machine 4 malfunctioned or encountered downtime, Part 3 production would cease entirely, posing a significant operational risk to the entire process. Further reducing system flexibility is the 12-hour setup time for Part 3 on Machine 4, which alone uses around 10% of the available weekly regular production capacity. It is highly advised that management look into the viability of cross-training Machine 2 or Machine 5 to generate Part 3 to reduce this danger.

Finding 1.2: Demand Sensitivity Reveals Capacity Ceiling

Demand Level	Total Overtime	Feasibility
80%	~45 hrs	Comfortable
100%	~85 hrs	Normal
110%	~120 hrs	Stressed
120%	~160 hrs	Near limit
130%	Infeasible	Cannot meet

Table 23: Demand Sensitivity Analysis

Discussion: FDC runs at around 70% of its maximum production capacity under typical demand conditions, which offers some flexibility for unexpected demand surges but leaves little room for variability. Even with maximum overtime, a 30% increase in demand would exceed the system's capacity, making it impossible to satisfy consumer demands on time. Establishing an early warning system is advised in order to proactively control this risk. In order to improve demand management and more realistic delivery commitments, management should promptly notify customers of the possibility of delays when incoming orders above 115% of the baseline demand level and determine whether overtime should be approved.

7.2: Quality Improvement ROI

Finding 2.1: Yield Improvement is the Highest ROI Investment

Yield Improvement	Overtime Reduction	Annual Savings*
+5% (all parts)	1215 hrs/week	\$18,00022,500
+10% (all parts)	2228 hrs/week	\$33,00042,000
+15% (all parts)	3038 hrs/week	\$45,00057,000

Table 24: Yield Sensitivity Analysis

The table shows that in order to boost manufacturing output across all components, a rapid increase in extra hours and associated costs is required. An additional \$18,000 to \$22,500 per week, or roughly 12 to 15 hours of overtime, would be required for a 5% increase in output. Requirements rise to 2228 overtime hours per week when output is increased by 10%, with total costs ranging from \$33,000 to \$42,000. The system significantly depends on overtime at a 15% increase, needing 3038 hours per week and raising expenses to between \$45,000 and \$57,000. This pattern shows that while small production increases can be handled with little overtime, higher demand levels eventually become more costly and unsustainable, underscoring the significance of capacity planning and productivity gains over long-term reliance on overtime.

Part	Current Yield	Improvement Value
Part 2	55%	\$4,200/year (highest)
Part 1	60%	\$3,800/year
Part 5	60%	\$3,500/year
Part 4	65%	\$2,900/year

Part 3	75%	\$1,800/year (lowest)
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Table 25: Part Specific Yield Impact

It is recommended that quality improvement efforts be prioritized for Part 2 because Parts 1 and 5 have the biggest potential return from yield improvements. In particular, a \$50,000 investment in improving Part 2 quality that increases yield by 5% would pay for itself in less than six months, making it a very cost-effective upgrade. Focusing on these areas first will maximize financial gain in addition to improving overall production efficiency and product reliability.

Finding 2.2: Setup Time Reduction Has Diminishing Returns

Setup Reduction	Overtime Reduction	Annual Savings
25%	812 hrs/week	\$12,00018,000
50%	1520 hrs/week	\$22,50030,000

Table 26: Setup Sensitivity Analysis

Key Insight

- Setup reduction is valuable but less impactful than yield improvement.
- Machine 3's 24 hour setup for Part 5 is the prime SMED target.

Discussion: After implementing yield improvement programs, investing in setup time reduction need to be the next highest priority. Focus should be given to the Part 5 setup time on Machine 3, which now takes 24 hours but may be cut in half to 12 hours. Reaching this reduction would increase production flexibility, free up a large amount of capacity, and reduce reliance on overtime, all of which would lead to more effective resource use and better responsiveness to demand fluctuations.

7.3: Cost Structure Optimization

Finding 3.1: Support Personnel Costs Drive Scheduling Philosophy

Cost Ratio	Optimal Strategy	Total OT	Max OT
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0.5	Concentrate OT	85 hrs	35 hrs
1.0	Hybrid	92 hrs	28 hrs
1.33 (current)	Transition point	98 hrs	22 hrs
2.0	Balance OT	110 hrs	22 hrs

Table 27: Cost Ratio Sensitivity Analysis

Key Insight

- Current costs: $c_{\text{prod}} = \$30/\text{hr}$, $c_{\text{supp}} = \$40/\text{hr} \rightarrow \text{Ratio} = 1.33$
- FDC is exactly at the transition point where solution structure changes

Discussion: With the current cost structure, a hybrid strategy is the best option. If support staff costs increase, move toward more balanced scheduling (new union contract, expert hiring).

Finding 3.2: The Balancing Cost is Quantifiable

Demand Level	Q2 Total OT	Q3 Total OT	Balancing Cost
90%	70 hrs	82 hrs	12 hrs (+17%)
100%	85 hrs	100 hrs	15 hrs (+18%)
110%	120 hrs	138 hrs	18 hrs (+15%)

Table 28: Q3 vs Q2 Comparison

Key Insights: The 1518% "balancing premium" provides: reduced worker fatigue, lower equipment wear, and better morale/retention.

7.4 Policy Recommendations

Finding 4.1: The No Inventory Policy is Costly

Holding Cost	Q7 OT	Q8 OT	OT Savings	Inventory
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\$1/unit	95 hrs	78 hrs	17 hrs	850 units
\$2/unit	95 hrs	82 hrs	13 hrs	520 units
\$3/unit	95 hrs	88 hrs	7 hrs	280 units
\$5/unit	95 hrs	93 hrs	2 hrs	80 units

Table 29: Inventory Policy Analysis (Q8 vs Q7)

Breakeven Analysis

- Inventory becomes worthwhile when holding cost < \$3.50/unit
- At \$2/unit holding cost: Annual savings = 13 hrs × \$30/hr × 50 weeks = \$19,500

Recommendation

It is advised to the company to loosen the current no-inventory policy if the holding cost is less than \$3 per unit. To mitigate the danger of production shortages and absorb demand spikes in Week 2, it would be beneficial to maintain a safety reserve of roughly 300500 units of strategic parts, particularly Parts 3 and 5. This strategy offers an efficient buffer against demand fluctuations while enhancing operational stability and service reliability at a comparatively modest cost.

Finding 4.2: Initial Setup Configuration Matters

Configuration	Total Overtime	Savings vs Q2
Config A (given)	78 hrs	7 hrs
Config B	75 hrs	10 hrs
Config C	82 hrs	3 hrs
Config D	85 hrs	0 hrs
Config E	80 hrs	5 hrs

Table 30: Setup Configuration Comparison

Optimal Week End Configuration (Config B)

Machine	End of Week Setup	Rationale
Machine 1	Part 4	High volume, saves 8hr setup
Machine 2	Part 1	Consistent demand
Machine 3	Part 5	Saves 24hr setup (critical!)
Machine 4	Part 3	Only machine for Part 3
Machine 5	Part 4	Backup for Part 4 surge

Table 31: Recommended Week End Setup

Recommendation

Since maintaining the Machine 3 → Part 5 configuration saves about 24 hours of production time, maintenance should be directed to give it top attention. To guarantee quicker starts, better capacity availability, and more seamless production flow at the start of every new week, Config B should also be used as the regular weekend configuration.

Finding 4.3: Shortfall Penalty Threshold

Penalty	Total Cost	Shortfall	Strategy
\$1/unit	\$3,200	2,100 units	Accept shortfalls
\$3/unit	\$4,800	850 units	Partial shortfall
\$5/unit	\$5,600	280 units	Minimize shortfalls
\$10/unit	\$6,400	0 units	100% delivery

Table 32: Penalty Sensitivity Analysis

Key Insight

- At penalties below \$5/unit, it's economically rational to accept some shortfalls
- The "all or nothing" transition occurs around \$78/unit penalty

7.5 Two-week Planning Benefits

Finding 5.1: Coordinated Planning Saves 815%

Week Pair	2×Q5 (Independent)	Q7 (Coordinated)	Savings
Weeks 12	165 hrs	148 hrs	17 hrs (10%)
Weeks 1112	195 hrs	168 hrs	27 hrs (14%)
Weeks 34	158 hrs	145 hrs	13 hrs (8%)

Table 33: Two Week vs Independent Planning

Key Insight

- Two week coordinated planning saves 1014% overtime
- Higher savings during high demand periods (Weeks 1112)

Recommendation

It is recommended to implement a two week rolling planning horizon to improve production coordination and decision making. As planners gain experience with this approach, the benefits will compound over time through better anticipation of demand, more effective capacity utilization, and the development of optimal carryover and scheduling patterns.

Finding 5.2: Demand Pattern Affects Strategy

Pattern	Total OT	Best Strategy
Week 1 > Week 2	142 hrs	Frontload production, carry over setups
Week 1 = Week 2	148 hrs	Balance across weeks
Week 1 < Week 2	155 hrs	Prepare Week 2 setups in Week 1

Table 34: Impact of Demand Patterns

This table shows how the dispersion of demand over subsequent weeks affects scheduling strategy and overtime. A descending schedule (Week 1 > Week 2) results in the lowest overtime at 142 hours by frontloading production and stretching setups into the lighter second week. To meet equal demand over several weeks, 148 hours of balanced production are required. The most challenging pattern at 155 hours is a rising pattern (Week 1 < Week 2), requiring Week 1 to set up for the more taxing Week 2. This explains why, due to rising demand, advance planning is required to avoid bottlenecks during FDC's Week 1011 switch.

7.6 Risk Analysis

Finding 6.1: Demand Volatility Risk

Scenario	Probability	Impact	Mitigation
20% demand spike	15%	+40 hrs OT, shortfalls	Preauthorize OT, communicate early
Machine 4 failure	5%	Part 3 production stops	Crosstrain backup machine
Yield drop 5%	10%	+15 hrs OT	Quality monitoring, root cause
Support staff shortage	10%	Can't run weekend OT	Crosstrain production staff

Table 35: Scenario Risk Analysis

A thorough risk evaluation of possible interruptions to FDC's production activities is shown in this table. Four major possibilities that could have a detrimental effect on production scheduling are identified by the investigation. Preauthorized overtime procedures and early customer communication can reduce the risk of delivery shortages caused by a 20% demand increase, which has a 15% chance of happening and would necessitate an additional 40 hours of overtime. Since Machine 4 is the only machine capable of creating Part 3, its failure would entirely stop production, even if its chance is only 5%. This makes Machine 4 failure the most significant operational risk. To fix this issue, operators must receive cross training on a backup machine. A 10% probability of a 5% yield reduction would increase weekly overtime by about 15 hours, highlighting the necessity of ongoing quality control and root cause analysis initiatives. Lastly, there is a 10% chance that a lack of support personnel will prevent weekend overtime operations; however, this risk can be reduced by cross training production workers to handle support tasks.

Finding 6.2: Sensitivity Rankings

Rank	Parameter	Impact on Cost	Controllable?
1	Demand	±30%	No (external)
2	Yield factors	±20%	Yes (quality)
3	Penalty cost	±25%	Partially (contracts)
4	Setup times	±15%	Yes (SMED)
5	Holding cost	±10%	Partially
6	Support cost ratio	±10%	No (labor market)

Table 36: Parameters Ranked by Solution Impact

This table gives management a framework for setting priorities for improvement projects by ranking the six most important factors influencing FDC's production costs. The largest influence ($\pm 30\%$ on total cost) is caused by demand unpredictability, which is mostly unpredictable due to its reliance on external market conditions. Yield factors rank second ($\pm 20\%$ impact) and represent the most attractive improvement opportunity since they are directly controllable through quality programs. Penalty costs ($\pm 25\%$ impact) can be partially influenced through customer contract negotiations. Setup times ($\pm 15\%$ impact) are fully controllable through SMED (Single Minute Exchange of Dies) initiatives, making them a strong candidate for process improvement investment. Holding costs and support cost ratios have moderate impacts ($\pm 10\%$ each) but offer limited controllability.

7.7 Implementation Roadmap

Quick Wins (03 months)

Action	Investment	Expected Savings	Payback
Implement Config B setup pattern	\$0	\$4,500/year	Immediate
Two week planning horizon	\$5,000	\$25,000/year	2 months
Early warning at 115% demand	\$0	Risk reduction	Immediate

Table 37: Quick Win Actions

This table lists three quick fixes that yield quick results with little initial investment. By maintaining appropriate machine configurations, especially the crucial 24hour Machine 3 → Part 5 setup, Config B's implementation as the typical weekend setup pattern saves \$4,500 yearly at no expense. Transitioning to a two week planning horizon requires just \$5,000 in training but delivers \$25,000 in annual savings with a 2month payback. Establishing an early warning system at 115% demand costs nothing and enables proactive customer communication before capacity constraints become critical.

Medium Term (3-12 months)

Action	Investment	Expected Savings	Payback
Part 2 quality improvement (5% yield)	\$50,000	\$21,000/year	2.4 years
Machine 3 setup reduction (SMED)	\$30,000	\$12,000/year	2.5 years
Inventory policy pilot	\$20,000 WC	\$15,000/year	1.3 years

Table 38: Medium Term Actions

This table outlines three strategic initiatives with 13 year payback periods. The Part 2 quality improvement program targets a 5% yield increase for the lowest yield part (55%), requiring \$50,000 but generating \$21,000 annually. Machine 3 setup reduction using SMED methodology targets the longest setup time (24 hours), with \$30,000 investment yielding \$12,000 annually. The inventory policy pilot requires \$20,000 in working capital to stock strategic parts, generating \$15,000 in annual savings with the fastest payback of 1.3 years.

Long Term (12+ months)

Action	Investment	Expected Savings	Payback
Crosstrain Machine 2 for Part 3	\$100,000	Risk reduction	Strategic
All parts yield improvement program	\$200,000	\$60,000/year	3.3 years
Integrated planning software	\$75,000	\$30,000/year	2.5 years

Table 39: Long Term Actions

Three strategic initiatives that offer substantial long term value but demand a large cash commitment are shown in this table. By reducing the single point of failure vulnerability where Machine 4 downtime stops all Part 3 production, cross training Machine 2 operators to create Part 3 involves a \$100,000 investment and offers strategic risk reduction rather than direct cost savings. Building on the lessons acquired during the medium term Part 2 trial, the all parts yield improvement program is a comprehensive \$200,000 endeavor aimed at quality improvements across all five parts, producing \$60,000 in yearly savings with a 3.3year payback. With a 2.5year payback period, integrated planning software (\$75,000 investment) would automate the two week rolling horizon optimization, systematically capture setup carryover opportunities, and produce \$30,000 in annual savings.

8.Future Work

8.1 Proposed Further Scenarios

The optimization models and sensitivity analysis developed in this project provide a solid foundation for FDC's production planning. However, several extensions would provide deeper insights and address real world complexities not captured in the current deterministic framework. This section proposes eight scenario extensions organized by strategic priority.

8.1.1 Dynamic optimization (Full 12Week Optimization)

Scenario Description

- We could tweak the setup to handle all 12 weeks at once. Instead of focusing on one week or splitting into two-week chunks, it'd run through a shifting frame - giving a clearer picture of where limits pop up and what demand looks like across the full stretch.

Why This Matters

- We use this model to help decide when to stock up during slow periods - say, week 9 or 10 - so we're ready for busier times. Since week 11 usually hits peak demand, making products earlier works better. That way, instead of rushing later, we stay ahead. It's about timing production just right before things get busy.
- Setting things up smarter: we found a two-week plan cut overtime by 10–14%. Going with four or twelve weeks might boost savings by another 58%, since setups can flow smoother one after another.

Implementation Approach

2. We could start off by guessing what the need might look like over the coming month.
3. Solve a weekly week plan each Monday morning - using fresh data helps adjust faster.
4. We can just try the Week 1 choices; then shift gears or pick a different path.
5. When fresh info shows up, changes need forecasts right away - so things stay on track without guesswork piling up.

8.1.2 Machine Breakdown and Preventive Maintenance Scenarios

Scenario Description

Picture times when gadgets fail entirely or barely work. Use that to figure out how much backup you need so things keep running smoothly - then plan fixes before problems hit.

Take Part 3 - only one weak spot matters. That part comes from just Machine 4. When Machine 4 fails, making Part 3 halts right away. Even if this happens only 5% of the time, delivery delays hit hard. So a small glitch causes big problems.

Scenario	Description	Probability	Impact	Mitigation to Test
Machine 4 Failure	Complete outage, 1 week	5%/year	Part 3 = 0	Crosstrain M2 for Part 3
Machine 3 Degradation	50% capacity, 2 weeks	10%/year	Part 5 constrained	Shift Part 5 to M5
Multi-Machine Event	2 machines at 70%	3%/year	Severe shortage	Emergency OT protocol
Planned Maintenance	1 machine offline/week	Scheduled	Capacity reduction	Optimal PM scheduling

Table 19: Scenarios to Model

8.2 New Data Requirements and Further Analysis

The scenarios proposed above require data beyond what was available for this project. This section details the data collection initiatives needed to support richer models, organized by priority and implementation difficulty.

8.2.1 Detailed Cost Data

Current Limitation

The models used simplified cost parameters: \$30/hr overtime, \$40/hr support, \$2/unit inventory, \$3/unit backlog. These may not reflect true costs, especially for differentiated penalty structures.

Data Required

Cost Element	Granularity	Source
Actual OT labor costs	By skill level/shift	Payroll
Support staff breakdown	By function (QC, main, material)	Accounting
Inventory carrying cost	By part (storage, insurance, obsolescence)	Finance

Customer penalty clauses	By part/customer	Contracts
Expediting/air freight costs	Historical	Logistics
Machine operating costs	By machine (energy, consumables)	Operations

Table 40: Data Required

What's FDC benefits from this?

- Real cost tracking: swap guesses with real numbers to compare choices better - so you see what things actually take from your wallet instead of hoping it's right.
- Part-specific penalties: the model adjusts service tiers using real contract details instead.
- Activity Based Costing: Allocate overhead costs to specific machinepart combinations.

8.2.3 Validation Through Simulation

Present Restrictions

Under idealized assumptions, optimization models offer the best answers. The dynamic interactions of actual production systems, such as queue buildups, operator variability, unforeseen events, and execution of flaws, are not captured by them.

Method of Simulation

Create a discrete event simulation (DES) model of the production system at FDC that can:

1. Take the optimized schedule as input
2. Adding stochastic unpredictability to yields, processing durations, and setup times
3. Simulate sporadic machine malfunctions and repair times
4. Monitor resource usage, queue times, and WIP inventories
5. Compare planned and real performance across hundreds of replications.